CSCI 3130 – Formal Languages and Automata Theory

Mock Midterm Examination

Covering Lectures 1–6 (Chapters 1–6)

Instructions: Answer all questions in the space provided. Total: 100 points. Time suggested: 120 minutes. Closed book, no calculators. Unless stated otherwise, $\Sigma = \{a, b\}$ means the alphabet is $\{a, b\}$ and λ denotes the empty string.

Question	Points
Q1. Mathematical Preliminaries & Proofs	10
Q2. Languages and Grammars	12
Q3. Deterministic Finite Accepters (DFA)	14
Q4. Nondeterministic Finite Accepters (NFA)	14
Q5. Regular Expressions & Regular Grammars	12
Q6. Closure Properties & Decision Questions	12
Q7. Pumping Lemma	14
Q8. Context-Free Grammars & Normal Forms	12
Total	**100**

Part I – Questions

Question 1 (Mathematical Preliminaries and Proofs – 10 points)

(a) (2 pts) Let $A = \{1, 2, 3\}$ and $B = \{2, 3, 4, 5\}$. Compute $A \cup B$, $A \cap B$, $A - B$, and the powerset 2^B restricted to subsets of size 2. How many subsets does 2^B have in total?

(b) (2 pts) Use DeMorgan's laws to simplify $\overline{(A \cap B) \cup C}$ into a form with no complement applied to a union or intersection, given a universe U. (Write the answer using complements of single sets only.)

(c) (3 pts) Prove by induction that a binary tree of height n has at most 2^n leaves. State the basis, the inductive hypothesis, and the inductive step explicitly.

(d) (3 pts) Prove by contradiction that $\sqrt{2}$ is not rational.

Solution 1

(a) $A \cup B = \{1, 2, 3, 4, 5\}$; $A \cap B = \{2, 3\}$; $A - B = \{1\}$ (and $B - A = \{4, 5\}$). The subsets of B of size 2 are $\{2,3\}, \{2,4\}, \{2,5\}, \{3,4\}, \{3,5\}, \{4,5\}$. Since $|B| = 4$, the powerset has $|2^B| = 2^{|B|} = 2^4 = 16$ subsets.

(b) By DeMorgan: $\overline{(A \cap B) \cup C} = \overline{A \cap B} \cap \bar{C} = (\bar{A} \cup \bar{B}) \cap \bar{C}$.

(c) **Theorem:** a binary tree of height n has at most 2^n leaves. Let $L(i)$ = maximum number of leaves of a subtree at height i. **Basis:** $L(0) = 1$ (the root alone). **Hypothesis:** assume $L(i) \le 2^i$ for all $i = 0, 1, \ldots, k$. **Step:** a subtree of height $k+1$ consists of a root with at most two children, each the root of a subtree of height at most k, so $L(k+1) \le 2 \cdot L(k) \le 2 \cdot 2^k = 2^{k+1}$. Done.

(d) Assume $\sqrt{2}$ is rational: $\sqrt{2} = n/m$ with n, m integers having no common factors. Then $2m^2 = n^2$, so n^2 is even, hence n is even, $n = 2k$. Then $2m^2 = 4k^2$, i.e. $m^2 = 2k^2$, so m^2 is even and hence m is even, $m = 2p$. Thus n and m share the factor 2, contradicting that they have no common factors. Therefore $\sqrt{2}$ is not rational. ■

Question 2 (Languages and Grammars – 12 points)

Let $\Sigma = \{a, b\}$ and let $L = \{a^n b^n : n \geq 0\}$.

(a) (2 pts) Is $\lambda \in L$? Is $aab \in L$? Is $aabbaaabbb \in L^2$? Justify briefly.

(b) (3 pts) Give a grammar G with $L(G) = L$ and show a derivation of $aabb$ in G. Identify one sentential form and one sentence in your derivation.

(c) (3 pts) Give a grammar for $L' = \{a^n b^{n+1} : n \geq 0\}$ and briefly explain why it works.

(d) (2 pts) Consider G'' : $S \to SS \mid \lambda \mid aSb \mid bSa$ over $\{a, b\}$. What language does G'' generate? (You do not need a full proof, but name the language.)

(e) (2 pts) When are two grammars G_1 and G_2 said to be *equivalent*? Give an example of two different (non-identical production sets) equivalent grammars.

Solution 2

(a) $\lambda = a^0 b^0$, so $\lambda \in L$. $aab \notin L$ (the numbers of a's and b's differ). $L^2 = \{a^n b^n a^m b^m : n, m \geq 0\}$, and with $n = 2$, $m = 3$: $aabbaaabbb = a^2 b^2 a^3 b^3 \in L^2$. Yes.

(b) $G = (\{S\}, \{a, b\}, S, P)$ with $P = \{S \to aSb,\ S \to \lambda\}$. Derivation: $S \Rightarrow aSb \Rightarrow aaSbb \Rightarrow aabb$. Sentential forms: S, aSb, $aaSbb$ (e.g. $aaSbb$ contains the variable S). Sentence: $aabb$ (all terminals).

(c) $G' = (\{S, A\}, \{a, b\}, S, P')$ with $S \to Ab$, $A \to aAb$, $A \to \lambda$. Applying $A \to aAb$ n times and then $A \to \lambda$ gives $a^n b^n$, and the final $S \to Ab$ appends one extra b: $L(G') = \{a^n b^{n+1} : n \geq 0\}$. Example: $S \Rightarrow Ab \Rightarrow aAbb \Rightarrow aabb$.

(d) $L(G'') = \{w \in \{a, b\}^* : n_a(w) = n_b(w)\}$ – all strings with the same number of a's and b's (proof by induction on $|w|$: if w starts and ends with different letters, peel them with aSb/bSa; otherwise split $w = w_1 w_2$ and use $S \to SS$).

(e) G_1 and G_2 are equivalent iff $L(G_1) = L(G_2)$. Example: G_1 : $S \to aSb \mid \lambda$ and G_2 : $S \to aAb \mid \lambda$, $A \to aAb \mid \lambda$; both generate $\{a^n b^n : n \geq 0\}$.

Question 3 (Deterministic Finite Accepters – 14 points)

(a) (4 pts) Design a DFA over $\Sigma = \{0, 1\}$ that accepts exactly the strings that contain the substring 001. Draw the state diagram.

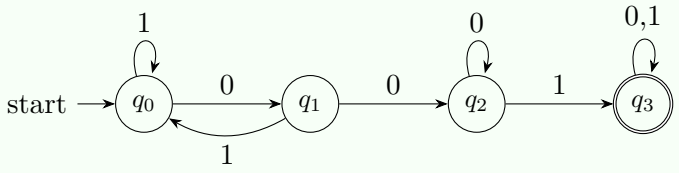
(b) (4 pts) Write the 5-tuple $M = (Q, \Sigma, \delta, q_0, F)$ for your DFA in (a) using a transition table.

(c) (3 pts) For a DFA, define the extended transition function δ^* and the language $L(M)$. Illustrate with $\delta^*(q_0, 00110)$ in your machine.

(d) (3 pts) Design a DFA for $L = \{w \in \{a, b\}^* : |w| \bmod 3 = 0\}$. How many states do you need? Why is this number minimal?

Solution 3

(a) States remember the longest prefix of 001 seen so far: q_0 (no progress), q_1 (seen 0), q_2 (seen 00), q_3 (seen 001, accepting trap, loops on 0, 1). Transitions: $q_0 \xrightarrow{0} q_1$, $q_0 \xrightarrow{1} q_0$; $q_1 \xrightarrow{0} q_2$, $q_1 \xrightarrow{1} q_0$; $q_2 \xrightarrow{0} q_2$, $q_2 \xrightarrow{1} q_3$; $q_3 \xrightarrow{0,1} q_3$.



(b) $M = (Q, \Sigma, \delta, q_0, F)$ with $Q = \{q_0, q_1, q_2, q_3\}$, $\Sigma = \{0, 1\}$, $F = \{q_3\}$:

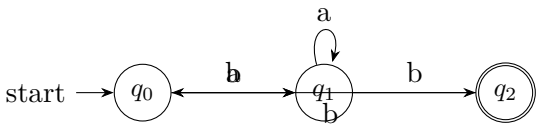
δ	0	1
q_0	q_1	q_0
q_1	q_2	q_0
q_2	q_2	q_3
q_3	q_3	q_3

(c) Recursive definition: $\delta^*(q, \lambda) = q$ and $\delta^*(q, wa) = \delta(\delta^*(q, w), a)$. Language: $L(M) = \{w \in \Sigma^* : \delta^*(q_0, w) \in F\}$. For $w = 00110$: $\delta^*(q_0, 0) = q_1$, $\delta^*(q_0, 00) = q_2$, $\delta^*(q_0, 001) = q_3$, and from q_3 the machine loops, so $\delta^*(q_0, 00110) = q_3 \in F$: 00110 is accepted (it contains the substring 001).

(d) Use a 3-state cycle counting $|w|$ mod 3: states $0, 1, 2$, start 0, final $\{0\}$, $\delta(i, a) = \delta(i, b) = (i + 1)$ mod 3. Three states are minimal: the states reached by λ, a, aa must be pairwise distinguishable. Reached by λ vs a: reading λ, the first ends final, the second does not. Reached by a vs aa: reading b, we reach state $|ab| = 2$ (non-final) vs state $|aab| = 3$ (final). Three pairwise distinguishable states force at least 3 states.

Question 4 (Nondeterministic Finite Accepters – 14 points)

Consider the NFA M with $\Sigma = \{a, b\}$:



(a) (3 pts) Give $\delta^*(q_0, a)$, $\delta^*(q_0, ab)$, and $\delta^*(q_0, aaabbb)$. Which of the strings ab, $aaabbb$, $abab$ are accepted?

(b) (3 pts) What is $L(M)$? Describe it in words and as a set or regular expression.

(c) (8 pts) Apply the subset construction to convert M into an equivalent DFA M'. Show all DFA states, the transition function, the initial state, and the final states. Verify that ab is accepted by M'.

Solution 4

(a) $\delta^*(q_0, a) = \{q_1\}$; $\delta^*(q_0, ab) = \{q_2\}$; for $aaabbb$: $q_0 \xrightarrow{a} q_1 \xrightarrow{a} q_1 \xrightarrow{a} q_1 \xrightarrow{b} q_2 \xrightarrow{b} q_0 \xrightarrow{b} q_1$, so $\delta^*(q_0, aaabbb) = \{q_1\}$. Accept: $ab \in L(M)$ since $\{q_2\}$ contains a final state; $aaabbb \notin L(M)$ (ends at $\{q_1\}$, not final); $abab$: after ab we are at q_2, but $\delta(q_2, a) = \emptyset$ – the only computation hangs, so $abab \notin L(M)$.

(b) From the DFA computed in (c): $L(M) = ((a \cup b)a^*bb)^*(a \cup b)a^*b$. In words: any number of "round trips": one letter, then any number of a's, then exactly two b's (the first b reaches the final state, the second returns to the start state), followed by a final block: one letter, any number of a's, and a single b ending in the final state. Check: ab: zero round trips plus final block ab – accepted. ✓

(c) Subset construction. NFA transitions: $\delta(q_0, a) = \delta(q_0, b) = \{q_1\}$; $\delta(q_1, a) = \{q_1\}$; $\delta(q_1, b) = \{q_2\}$; $\delta(q_2, b) = \{q_0\}$; all others $\emptyset$. DFA M': Start $A = \{q_0\}$. $\delta'(A, a) = \{q_1\} = B$; $\delta'(A, b) = \{q_1\} = B$. $B = \{q_1\}$: $\delta'(B, a) = \{q_1\} = B$; $\delta'(B, b) = \{q_2\} = C$. $C = \{q_2\}$: $\delta'(C, a) = \emptyset = D$; $\delta'(C, b) = \{q_0\} = A$. $D = \emptyset$: loops on a, b. Final states of M': states containing q_2, i.e. $F' = \{C\}$. ab: $A \xrightarrow{a} B \xrightarrow{b} C \in F'$: accepted. ✓(This DFA confirms the language in (b): loops $A \to B \xrightarrow{a^*} B \xrightarrow{b} C \xrightarrow{b} A$ plus the exit $A \to B \xrightarrow{a^*} B \xrightarrow{b} C$.)

Question 5 (Regular Expressions and Regular Grammars – 12 points)

(a) (3 pts) Give a regular expression over $\Sigma = \{0, 1\}$ for $L_1 = \{w : w$ contains at least one pair of consecutive 0s$\}$.
(b) (3 pts) Give a regular expression for $L_2 = \{w \in \{0, 1\}^* : w$ does not end in 01$\}$.
(c) (3 pts) Give a regular expression for $L_3 = \{w \in \{a, b\}^* : n_a(w)$ is even$\}$.
(d) (3 pts) Consider the right-linear grammar G : $V_0 \to aV_1$, $V_1 \to abV_0 \mid b$. Construct an NFA that accepts $L(G)$ and give $L(G)$ as a regular expression.

Solution 5

(a) $r_1 = (0 + 1)^*00(0 + 1)^*$ (anything, then 00, then anything).
(b) If x does not end in 01, then $|x| < 2$ or x ends in 00, 10, or 11: $r_2 = \lambda + 0 + 1 + (0 + 1)^*(00 + 10 + 11)$.
(c) Express x as blocks around each odd occurrence pattern: an even number of a's means: some b's, then pairs of a's separated by arbitrary b's: $r_3 = b^*(ab^*ab^*)^*$, equivalently $(b^*ab^*a)^*b^*$ or $(b + ab^*a)^*$.
(d) From G: variables V_0, V_1 become NFA states, plus final state V_F; start V_0. $V_0 \to aV_1$: edge $V_0 \xrightarrow{a} V_1$. $V_1 \to abV_0$: chain $V_1 \xrightarrow{a} X \xrightarrow{b} V_0$. $V_1 \to b$: edge $V_1 \xrightarrow{b} V_F$ (V_F final). $L(G) = (aab)^*ab$ (the loop $a\,a\,b$ returns to V_0; the exit $a\,b$ leads to the final state).

Question 6 (Closure Properties and Decision Questions – 12 points)

(a) (3 pts) Let $L_1 = \{a^nb : n \geq 0\}$ and $L_2 = \{ba\}$, both regular. Using only the standard NFA constructions (union, concatenation, star, reversal), build an NFA for $L_1^R \cup L_1L_2$. Sketch the construction (state diagrams) and state what language the result accepts.
(b) (4 pts) For DFAs $M_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$ and $M_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$, write down the product DFA M that accepts $L(M_1) - L(M_2)$, giving Q, δ, q_0, and F. (Note: for the complement construction, a DFA must be complete – add a trap state if some transitions are missing.)
(c) (2 pts) Given a regular language L (by its DFA), how do you decide each of the following: (i) $w \in L$; (ii) $L = \emptyset$; (iii) L is infinite.
(d) (3 pts) Let $h : \{a, b\} \to \{a, b, c\}^*$ be the homomorphism with $h(a) = ab$, $h(b) = bbc$. Compute $h(aba)$ and the homomorphic image of the language denoted by $r = (a+b^*)(aa)^*$.

Solution 6

(a) $L_1 = \{a^nb : n \geq 0\}$ has NFA M_1: $q_0 \overset{a}{\circlearrowleft} \xrightarrow{b} q_f$. $L_1^R = \{ba^n : n \geq 0\}$: reverse all transitions and swap initial/final: q_f becomes initial: $q_f \xrightarrow{b} q_0$, $q_0 \overset{a}{\circlearrowleft}$, with q_0 final (and add a new single final state if needed). $L_1L_2 = \{a^nbba : n \geq 0\}$: concatenate M_1 with M_2 (M_2: $p_0 \xrightarrow{b} p_1 \xrightarrow{a} p_2$) via a λ-edge from q_f to p_0, final state p_2. Then $L_1^R \cup L_1L_2$: new start with λ-edges into both machines and a single new final state. Result: $\{ba^n : n \geq 0\} \cup \{a^nbba : n \geq 0\}$.
(b) $M = (Q_1 \times Q_2, \Sigma, \delta, (q_1, q_2), F)$ where $\delta((p, q), a) = (\delta_1(p, a), \delta_2(q, a))$ and $F = \{(p, q) : p \in F_1$ and $q \notin F_2\}$. (Equivalently $L_1 \cap \bar{L}_2$.)
(c) (i) Simulate the DFA on w; accept iff the final state $\delta^*(q_0, w) \in F$. (ii) $L = \emptyset$ iff no final state is reachable from q_0 (graph reachability). (iii) L is infinite iff some final state is reachable from q_0 via a walk containing a cycle; otherwise finite.
(d) $h(aba) = h(a)h(b)h(a) = ab\,bbc\,ab = abbbcab$. The language denoted by $r = (a + b^*)(aa)^*$ is $L(r) = \{a, b, b^2, b^3, \ldots\}\{a^{2n} : n \geq 0\}$. By Theorem 4.3 the homomorphic image is regular and is denoted by substituting the images of the letters into r: $h(r) = (ab + (bbc)^*)(abab)^*$.

Question 7 (Pumping Lemma – 14 points)

(a) (2 pts) State the Pumping Lemma for regular languages, naming all three conditions on the decomposition $w = xyz$.

(b) (6 pts) Prove, using the Pumping Lemma, that $L = \{a^n b^n : n \geq 0\}$ is not regular. Structure your proof as a game: what does the opponent give, what string do you pick, and what contradiction follows for *every* possible decomposition?

(c) (6 pts) Prove that $L = \{a^p : p \text{ is prime}\}$ over $\Sigma = \{a\}$ is not regular.

Solution 7

(a) If L is an infinite regular language, then there exists an integer m such that every $w \in L$ with $|w| \geq m$ can be written $w = xyz$ with (1) $|xy| \leq m$, (2) $|y| \geq 1$, and (3) $xy^i z \in L$ for all $i = 0, 1, 2, \ldots$

(b) Assume L is regular. Let m be the pumping constant. Pick $w = a^m b^m \in L$; $|w| = 2m \geq m$. Consider any decomposition $w = xyz$ with $|xy| \leq m$ and $|y| \geq 1$. Since $|xy| \leq m$, both x and y lie entirely in the a-part: $y = a^k$ for some $k \geq 1$. Then $xy^2z = a^{m+k}b^m$. Since $k \geq 1$, $m + k \neq m$, so $a^{m+k}b^m \notin L$, contradicting condition (3). Hence no such decomposition works and L is not regular. ■ (Game form: O gives m; P picks $a^m b^m$; O chooses $y = a^k$, $1 \leq k \leq m$; P pumps $i = 2$ and wins.)

(c) Assume L regular with constant m. Choose $p =$ the smallest prime $\geq m$, and pick $w = a^p \in L$. Any decomposition with $|xy| \leq m$ gives $y = a^k$, $1 \leq k \leq m$ (with $k \leq p$). Pump $i = p + 1$: $xy^{p+1}z = a^{p+pk} = a^{p(k+1)}$. Since $k \geq 1$, $p(k + 1)$ is composite, so $xy^{p+1}z \notin L$ – contradiction. Hence L is not regular. ■ (Equivalently, pump with $i - 1 = p$: length $p + (i - 1)k = p(k + 1)$.)

Question 8 (Context-Free Grammars and Normal Forms – 12 points)

(a) (2 pts) Give a CFG for $L = \{ww^R : w \in \{a, b\}^*\}$ and show a derivation of $abba$.

(b) (3 pts) Show that $E \to E + E \mid E * E \mid (E) \mid a$ is ambiguous by exhibiting two different leftmost derivations of $a + a * a$. Then write an equivalent unambiguous grammar.

(c) (3 pts) Convert G: $S \to ABa$, $A \to aab$, $B \to Ac$ into Chomsky normal form, showing the intermediate steps (terminal replacement and binarization).

(d) (4 pts) Apply the CYK algorithm to determine whether $w = aabbb$ is in $L(G)$ for the CNF grammar $S \to AB$, $A \to BB \mid a$, $B \to AB \mid b$. Show the dynamic programming table (the triangle of substring sets).

Solution 8

(a) G: $S \to aSa \mid bSb \mid \lambda$. Derivation: $S \Rightarrow aSa \Rightarrow abSba \Rightarrow abba$ (then $S \Rightarrow \lambda$). $abba = (ab)(ab)^R$? Indeed $abba$ with $w = ab$: $w^R = ba$, $ww^R = abba$. ✓

(b) Leftmost derivation 1: $E \Rightarrow E + E \Rightarrow a + E \Rightarrow a + E * E \Rightarrow a + a * E \Rightarrow a + a * a$. Leftmost derivation 2: $E \Rightarrow E * E \Rightarrow E + E * E \Rightarrow a + E * E \Rightarrow a + a * E \Rightarrow a + a * a$. Two distinct leftmost derivations ⇒ ambiguous. Unambiguous grammar (precedence: $*$ binds tighter than $+$): $E \to E + T \mid T$; $T \to T * F \mid F$; $F \to (E) \mid a$.

(c) Step 1 (terminals): add $T_a \to a$, $T_b \to b$, $T_c \to c$; replace terminals in long productions: $S \to ABT_a$, $A \to T_aT_aT_b$, $B \to AT_c$, plus $T_a \to a, T_b \to b, T_c \to c$. Step 2 (binarize): $S \to ABT_a \Rightarrow S \to AV_1$, $V_1 \to BT_a$. $A \to T_aT_aT_b \Rightarrow A \to T_aV_2$, $V_2 \to T_aT_b$. $B \to AT_c$ is already in CNF. Final CNF grammar: $S \to AV_1$; $V_1 \to BT_a$; $A \to T_aV_2$; $V_2 \to T_aT_b$; $B \to AT_c$; $T_a \to a$; $T_b \to b$; $T_c \to c$.

(d) CNF: $S \to AB$, $A \to BB|a$, $B \to AB|b$; $w = aabbb$. Table V_{ij} (variables deriving $w_i \cdots w_j$):

length	substrings				
1	$a \mapsto \{A\}$	$a \mapsto \{A\}$	$b \mapsto \{B\}$	$b \mapsto \{B\}$	$b \mapsto \{B\}$
2	$aa \mapsto \emptyset$	$ab \mapsto \{S, B\}$	$bb \mapsto \{A\}$	$bb \mapsto \{A\}$	
3	$aab \mapsto \{S, B\}$	$abb \mapsto \{A\}$	$bbb \mapsto \{S, B\}$		
4	$aabb \mapsto \{A\}$	$abbb \mapsto \{S, B\}$			
5	$aabbb \mapsto \{S, B\}$				

Example: $ab \mapsto \{S, B\}$ since $S \to AB$ and $B \to AB$ with $A \in V_{11}$ and $B \in V_{22}$ ($a \mapsto A$, $b \mapsto B$). Since $S \in V_{15}$, $aabbb \in L(G)$. ✓

End of Examination